$$a(t) = \frac{dv}{dt} \Rightarrow \int_{v_0}^{v(t)} dv = \int_{t_0}^{t} a(t)\,dt$$

$$v_0 = v(t_0)$$

$$\boxed{v(t) = v_0 + \int_{t_0}^{t} a(t)\,dt}$$

$$a(t) = \text{costante} = a \Rightarrow \boxed{v(t) = v_0 + a(t - t_0)}$$

moto rettilineo uniformemente accelerato

$$v(t) = \frac{dx}{dt} \Rightarrow x(t) = x_0 + \int_{t_0}^{t} v(t)\,dt$$

$$x(t) = x_0 + \int_{t_0}^{t} \left[v_0 + \int_{t_0}^{t} a(t)\,dt \right] dt =$$

$$= x_0 + \int_{t_0}^{t} v_0\,dt + \int_{t_0}^{t} \left[\int_{t_0}^{t} a(t)\,dt \right] dt$$

$$\Rightarrow \boxed{x(t) = x_0 + v_0 (t - t_0) + \int_{t_0}^{t} \left[\int_{t_0}^{t} a(t)\, dt \right] dt}$$

$$a(t) = \text{const} = a \quad \Rightarrow$$

$$x(t) = x_0 + v_0 (t - t_0) + a \int_{t_0}^{t} \left[\int_{t_0}^{t} dt \right] dt =$$

$$= x_0 + v_0 (t - t_0) + a \int_{t_0}^{t} (t - t_0)\, dt = \qquad t^* = t - t_0 \quad dt^* = dt$$

$$= x_0 + v_0 (t - t_0) + a \int_{0}^{t - t_0} t^*\, dt^* =$$

$$= x_0 + v_0 (t - t_0) + a \left[\frac{1}{2} t^{*2} \right]_0^{t - t_0} =$$

$$\boxed{x(t) = x_0 + v_0 (t - t_0) + \frac{1}{2} a (t - t_0)^2}$$

$$t_0 = 0 \Rightarrow \boxed{x(t) = x_0 + v_0 t + \frac{1}{2} a t^2}$$

$$v(x) \quad , \quad a(x) \qquad\qquad a = \frac{dv}{dt} \qquad x = x(t)$$

$$v[x(t)] \quad , \quad a[x(t)]$$

$$w = -x^2 \implies f(x) = e^{w} = f(w)$$

$$f(x) = e^{-x^2} \qquad \frac{df}{dx} = f'(x) = \frac{df}{dw}\frac{dw}{dx} = \frac{d}{dw}\left(e^{w}\right)\frac{d}{dx}\left(-x^2\right) =$$

$$= e^{w}(-2x) = -2x\, e^{-x^2}$$

$$g[f(x)] \qquad \boxed{\frac{dg}{dx} = \frac{dg}{df}\frac{df}{dx}}$$

$$a(x) = \frac{d\,v(x)}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

$$\int_{x_0}^{x} a(x)\,dx = \int_{v_0}^{v(x)} v\,dv \qquad\qquad v_0 = v(x_0)$$

$$\implies \frac{1}{2}v^2\Big|_{v_0}^{v(x)} = \int_{x_0}^{x} a(x)\,dx$$

$$\implies \frac{1}{2}\left(v^2(x) - v_0^2\right) = \int_{x_0}^{x} a(x)\,dx$$

$$\Rightarrow \boxed{v^2(x) = v_0^2 + 2\int_{x_0}^{x} a(x)\,dx}$$

$a(x) = \text{cost} = a$

$$\boxed{v^2(x) = v_0^2 + 2a(x - x_0)}$$

$[v] = m/s$ $\qquad$ $[a] = m/s^2$